VIT
Vellore Institute of Technology
CHENNAI

Reg. Number [redacted]

# Continuous Assessment Test (CAT) – I- January 2025

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| Programme | : | B. Tech CSE - Specialization in Data Science | Semester | : | Winter 24-25 |
| Course Code & Course Title | : | BCSE206L & Foundations of Data Science | Class Number | : | CH2024250501774 |
| Faculty | : | Dr. Nachiyappan S | Slot | : | F1+TF1 |
| Duration | : | 1.30 Hrs | Max. Mark | : | 50 |

| Q. No. | Question Text | Marks |
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| 1. | a) A retail company wants to improve customer retention. How can data science help in identifying key factors contributing to customer churn prediction? Assume your own dataset for the purpose. (4 Marks)<br><br>b) A healthcare organization is overwhelmed with patient records. Suggest ways in which data science can assist in predictive healthcare. Assume your own dataset for the purpose. (4 Marks)<br><br>c)List out the suitable tools for the above use cases.(2 Marks) | 5<br><br>5 |
| 2. | Differentiate between a data scientist and a data analyst's role in a company launching a recommendation system. Assume you are playing both roles in the company and justify your answer. | 10 |
| 3 | You are hired as a data scientist by a food delivery company facing declining customer satisfaction ratings. The management suspects the issue is related to delayed deliveries but lacks concrete insights. They want you to use data science to identify the root causes, predict delays, and recommend solutions to improve customer satisfaction.<br>Think through each stage of the data science process and apply the knowledge to solve the above realistic, business-critical problem. | 10 |
| 4. | You work for a movie streaming platform and are tasked with analysing user behaviour and movie performance. The database contains the following tables:<br><br>1. **Users Table**: Contains user_id, name, age, and subscription_start_date.<br>2. **Movies Table**: Contains movie_id, title, genre, and release_date.<br>3. **Ratings Table**: Contains rating_id, user_id, movie_id, rating (1-5), and rating_date.<br><br>a) Identify all movies released after January 1, 2020, with an average user rating of at least 4.<br>b) Combine the **Movies** and **Ratings** tables to display each movie's title, genre, and its total number of ratings.<br>c) Calculate the average rating for each genre and sort the results to identify the top-performing genres. | |

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| | d) For each movie, compute the movie's average rating and compare it to the average rating of all movies in the same genre using a window function. | 10 |
| 5. | You are a data analyst for a mid-sized **retail chain** that operates both physical stores and an e-commerce platform. Recently, management has noticed the following issues:<br>1. Declining in-store sales.<br>2. Higher cart abandonment rates on the website.<br>3. Customer complaints about delayed product deliveries.<br>Management has provided access to the following datasets:<br>1. **Sales Data**: Contains transaction details (store_id, product_id, quantity_sold, price, transaction_date).<br>2. **Customer Data**: Includes customer demographics (age, gender, location) and purchase history.<br>3. **Website Data**: Contains session details (session_id, user_id, pages_visited, time_spent, cart_abandonment).<br>4. **Logistics Data**: Tracks delivery times (order_id, shipment_date, delivery_date, delay_reason).<br>Based on your analysis, provide actionable recommendations for:<br>a. Improving in-store sales.<br>b. Reducing cart abandonment rates.<br>c. Minimizing delivery delays and enhancing customer satisfaction. | 10 |

********All the best**********